Chapter 6: Infinite Horizon Again

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6.1 Intertemporal Utility Maximization

- The chapter revisits the continuous-time infinite-horizon problem with dynamic programming.
- The economic setup is close to the Hamiltonian examples; the purpose is to compare solution methods.

Core idea

A path-choice problem is reduced to an instantaneous control problem plus the value of behaving optimally in the future.

6.1.1 The Setup

- The individual chooses the consumption path $\{c(\tau)\}_{\tau \geq t}$.
- Prices and factor rewards are taken as given.

$$U(t) = \int_t^{\infty} e^{-\rho[\tau - t]} u(c(\tau)) \, \mathrm{d}\tau. \tag{6.1.1}$$

$$\dot{a} = ra + w - pc. \tag{6.1.2}$$

6.1.2 Solving by Dynamic Programming: Value Function

- The value function gives the maximum utility attainable from current wealth $a(t)$.
- Under optimal behavior from t onward, overall utility equals the value function.

$$V(a(t)) \equiv \max_{\{c(\tau)\}_{\tau \geq t}} U(t) \quad \text{subject to} \quad \dot{a} = ra + w - pc.$$

If the household behaves optimally, $\quad U(t) = V(a(t))$.

6.1.2 Solving by Dynamic Programming: Bellman Prelude

- Construct "tomorrow" as $t + \Delta t$.
- Use additive separability of intertemporal utility.

$$U(t) = \int_t^{t+\Delta t} e^{-\rho[\tau - t]} u(c(\tau))\, \mathrm{d}\tau + \int_{t+\Delta t}^{\infty} e^{-\rho[\tau - t]} u(c(\tau))\, \mathrm{d}\tau.$$

$$\int_t^{t+\Delta t} e^{-\rho[\tau - t]} u(c(\tau))\, \mathrm{d}\tau \approx u(c(t))\Delta t, \qquad e^{-\rho \Delta t} \approx \frac{1}{1 + \rho \Delta t}.$$

6.1.2 Solving by Dynamic Programming: Bellman Equation

- Assume optimal behavior from $t + \Delta t$ onward.
- The whole future path is replaced by the future value $V(a(t + \Delta t))$.

$$V(a(t)) = \max_{c(t)} \left\{ u(c(t))\Delta t + \frac{1}{1 + \rho \Delta t} V(a(t + \Delta t)) \right\}.$$

$$\rho V(a(t)) = \max_{c(t)} \left\{ u(c(t)) + \frac{\mathrm{d}V(a(t))}{\mathrm{d}t} \right\}. \tag{6.1.3}$$

6.1.2 Solving by Dynamic Programming: Derivation

- Multiply by $1 + \rho\Delta t$, divide by Δt, then take $\Delta t \to 0$.

$$
\begin{aligned}
(1 + \rho\Delta t)V(a(t)) &= \max_{c(t)}\{(1 + \rho\Delta t)u(c(t))\Delta t + V(a(t + \Delta t))\}, \\
\rho V(a(t)) &= \max_{c(t)}\left\{(1 + \rho\Delta t)u(c(t)) + \frac{V(a(t + \Delta t)) - V(a(t))}{\Delta t}\right\}, \\
\rho V(a(t)) &= \max_{c(t)}\left\{u(c(t)) + \frac{\mathrm{d}V(a(t))}{\mathrm{d}t}\right\}.
\end{aligned}
$$

6.1.2 Solving by Dynamic Programming: Intuition

- The left-hand side is the utility-income flow from owning value $V(a(t))$.
- The right-hand side is current utility plus the capital gain in the value of optimal behavior.

$$\rho V(a(t)) = u(c(t)) + \dot{V}(a(t)).$$

- This parallels the no-arbitrage condition for an asset paying dividends:

$$r(t)v(t) = \pi(t) + \dot{v}(t).$$

6.1.2 Solving by Dynamic Programming: Alternative Derivation

- Differentiate overall utility $U(t)$ with respect to time.
- The lower limit removes current utility; future utility becomes closer in time.

$$
\begin{aligned}
U(t) &= \int_t^\infty e^{-\rho[\tau - t]} u(c(\tau))\,\mathrm{d}\tau, \\
\dot{U}(t) &= -u(c(t)) + \int_t^\infty \frac{\partial}{\partial t}\left(e^{-\rho[\tau - t]}\right) u(c(\tau))\,\mathrm{d}\tau \\
&= -u(c(t)) + \rho U(t).
\end{aligned}
$$

$$
\rho U(t) = u(c(t)) + \dot{U}(t), \qquad \rho V(a(t)) = u(c(t)) + \dot{V}(a(t)).
$$

6.1.2 Solving by Dynamic Programming: DP1

- Use $\mathrm{d}V(a(t))/\mathrm{d}t = V'(a(t))\dot{a}$.
- Substitute the budget constraint into the Bellman equation.

$$\rho V(a(t)) = \max_{c(t)}\{u(c(t)) + V'(a(t))[ra + w - pc]\}. \tag{6.1.4}$$

$$u'(c(t)) = pV'(a(t)). \tag{6.1.5}$$

6.1.2 Solving by Dynamic Programming: FOC

- Consumption is a function of the state variable: $c(t) = c(a(t))$.
- More consumption gives current marginal utility but reduces wealth by p units.

$$\underbrace{u'(c(t))}_{\text{marginal utility gain}} = \underbrace{pV'(a(t))}_{\text{value loss from lower wealth}} .$$

One extra unit of $c(t)$: $\Delta U \approx u'(c(t)), \qquad \Delta V \approx pV'(a(t))$.

6.1.2 Solving by Dynamic Programming: DP2

- Start from the maximized Bellman equation.
- Differentiate with respect to a and use the envelope theorem.

$$\rho V(a) = u(c(a)) + V'(a)[ra + w - pc(a)].$$

$$\begin{aligned} \rho V'(a) &= V''(a)[ra + w - pc] + V'(a)r, \\ (\rho - r)V'(a) &= V''(a)[ra + w - pc]. \end{aligned} \tag{6.1.6}$$

6.1.2 Solving by Dynamic Programming: Costate Evolution

- Compute the time derivative of the shadow value of wealth.
- Use $\dot{a} = ra + w - pc$ and equation (6.1.6).

$$\begin{aligned}\frac{\mathrm{d}V'(a)}{\mathrm{d}t} &= V''(a)\dot{a} = V''(a)[ra + w - pc] \\ &= (\rho - r)V'(a).\end{aligned}$$

$$\frac{\dot{V}'(a)}{V'(a)} = \rho - r. \tag{6.1.7}$$

6.1.2 Solving by Dynamic Programming: DP3

- Log-differentiate the FOC $u'(c) = pV'(a)$.
- Insert the costate evolution equation.

$$\begin{aligned}
\frac{u''(c)}{u'(c)}\dot{c} &= \frac{\dot{p}}{p} + \frac{\dot{V}'(a)}{V'(a)}, \\
\frac{u''(c)}{u'(c)}\dot{c} &= \frac{\dot{p}}{p} + \rho - r, \\
-\frac{u''(c)}{u'(c)}\dot{c} &= r - \frac{\dot{p}}{p} - \rho.
\end{aligned}$$

Keynes-Ramsey rule

Consumption growth is governed by the real return $r - \dot{p}/p$ relative to time preference ρ.

6.2 Comparing Dynamic Programming to Hamiltonians

- Dynamic programming and Hamiltonian methods deliver the same optimal path.
- Hence their optimality conditions coincide when the maximization problem is the same.

$$\text{Dynamic programming:} \qquad \frac{\dot{V}'(a)}{V'(a)} = \rho - r,$$

$$\text{Current-value Hamiltonian:} \qquad \frac{\dot{\lambda}}{\lambda} = \rho - r.$$

$$u'(c) = pV'(a), \qquad u'(c) = p\lambda.$$

6.2 Comparing Dynamic Programming to Hamiltonians: Costate

- The derivative of the value function equals the current-value costate variable.
- Economically, both measure the value of one additional unit of the state variable.

$$V'(a) = \lambda.$$

$$\lambda = \frac{\partial \text{value of the optimal program}}{\partial a}.$$

- This parallels the Lagrange multiplier in static constrained maximization.

6.3 Dynamic Programming with Two State Variables

- The method extends directly to multiple state variables.
- Example: wealth a and human capital h.

$$U(t) = \int_t^\infty e^{-\rho[\tau - t]} u(c(\tau))\, \mathrm{d}\tau.$$

$$\dot{a} = f(a, h, c), \qquad \dot{h} = g(a, h, c). \tag{6.3.1}$$

- Assume $f(\cdot)$ and $g(\cdot)$ increase in a, h and decrease in c.

6.3 Dynamic Programming with Two State Variables: DP1

- The Bellman equation now contains two marginal value terms.
- Each shadow value multiplies the law of motion of its state variable.

$$\begin{aligned}\rho V(a,h) &= \max_c \left\{ u(c) + \frac{\mathrm{d}V(a,h)}{\mathrm{d}t} \right\} \\ &= \max_c \{ u(c) + V_a f(\cdot) + V_h g(\cdot) \}.\end{aligned}$$

6.3 Dynamic Programming with Two State Variables: FOC

- There is one control variable c, hence one first-order condition.
- Consumption affects both wealth accumulation and human-capital accumulation.

$$u'(c) + V_a \frac{\partial f(\cdot)}{\partial c} + V_h \frac{\partial g(\cdot)}{\partial c} = 0. \tag{6.3.2}$$

$$\underbrace{u'(c) + V_a f_c}_{\text{one-state-variable terms}} + \underbrace{V_h g_c}_{\text{new human-capital effect}} = 0.$$

6.3 Dynamic Programming with Two State Variables: DP2

- There are two costate variables: V_a and V_h.
- Differentiate the maximized Bellman equation with respect to a and h.

$$\rho V_a = V_{aa} f(\cdot) + V_a \frac{\partial f(\cdot)}{\partial a} + V_{ha} g(\cdot) + V_h \frac{\partial g(\cdot)}{\partial a}, \tag{6.3.3}$$

$$\rho V_h = V_{ah} f(\cdot) + V_a \frac{\partial f(\cdot)}{\partial h} + V_{hh} g(\cdot) + V_h \frac{\partial g(\cdot)}{\partial h}. \tag{6.3.4}$$

6.3 Dynamic Programming with Two State Variables: Time Derivatives

- Compute time derivatives of the two costate variables.
- Insert the state equations $\dot{a} = f(\cdot)$, $\dot{h} = g(\cdot)$.

$$\frac{\mathrm{d}V_a(a,h)}{\mathrm{d}t} = V_{aa}\dot{a} + V_{ah}\dot{h} = V_{aa}f(\cdot) + V_{ah}g(\cdot),$$
$$\frac{\mathrm{d}V_h(a,h)}{\mathrm{d}t} = V_{ha}\dot{a} + V_{hh}\dot{h} = V_{ha}f(\cdot) + V_{hh}g(\cdot).$$

6.3 Dynamic Programming with Two State Variables: Costate Evolution

- Insert (6.3.3) and (6.3.4) into the time derivatives.
- Cross derivatives disappear using $V_{ah} = V_{ha}$.

$$\frac{\mathrm{d}V_a(a,h)}{\mathrm{d}t} = \rho V_a - V_a \frac{\partial f(\cdot)}{\partial a} - V_h \frac{\partial g(\cdot)}{\partial a},$$
$$\frac{\mathrm{d}V_h(a,h)}{\mathrm{d}t} = \rho V_h - V_a \frac{\partial f(\cdot)}{\partial h} - V_h \frac{\partial g(\cdot)}{\partial h}.$$

6.3 Dynamic Programming with Two State Variables: Ratios

- The evolution equations can be written as growth rates of shadow values.
- Relative shadow prices enter because each state can affect both laws of motion.

$$\frac{\dot{V}_a}{V_a} = \rho - \frac{\partial f(\cdot)}{\partial a} - \frac{V_h}{V_a}\frac{\partial g(\cdot)}{\partial a},$$

$$\frac{\dot{V}_h}{V_h} = \rho - \frac{V_a}{V_h}\frac{\partial f(\cdot)}{\partial h} - \frac{\partial g(\cdot)}{\partial h}.$$

One-state analogue: $\dfrac{\dot{V}'(a)}{V'(a)} = \rho - r.$

6.3 Dynamic Programming with Two State Variables: DP3

- Differentiate the FOC (6.3.2) with respect to time.
- Replace $\dot{V}_a$ and $\dot{V}_h$ using the costate equations from DP2.
- The solution principle is unchanged relative to the one-state case.

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[u'(c) + V_a f_c(\cdot) + V_h g_c(\cdot)\right] = 0.$$

Main difference

In general, the shadow prices cannot be eliminated from the resulting control equation without additional assumptions on $f(\cdot)$ or $g(\cdot)$.

6.4 Nominal and Real Interest Rates and Inflation

- The model illustrates nominal interest rates, real interest rates, and inflation.
- It also provides another decentralized economy solved through dynamic programming and reduced-form equilibrium conditions.

Purpose

Start from agents' behavior, impose market clearing, and obtain reduced-form relationships for real and nominal variables.

6.4.1 Firms, the Central Bank and the Government: Firms

- Firms operate a neoclassical technology under perfect competition.
- Factor prices equal value marginal products.

$$Y = Y(K, L).$$

$$w^K = p\frac{\partial Y}{\partial K}, \qquad w^L = p\frac{\partial Y}{\partial L}. \tag{6.4.1}$$

6.4.1 Firms, the Central Bank and the Government: Budgets

- The central bank issues money $M(t)$ in exchange for government bonds $B(t)$.
- The government finances spending and interest payments using taxes and debt issuance.

$$\begin{aligned} \text{Central bank:} \quad & iB + \dot{M} = \dot{B}, \\ \text{Government:} \quad & G + iB = T + \dot{B}. \end{aligned}$$

If only the central bank holds government bonds: $\dot{M} = G - T$.

6.4.2 Households

- Households obtain utility from consumption and real money balances.
- Total wealth equals capital goods plus money: $a(t) = k(t) + m(t)$.

$$\int_t^{\infty} e^{-\rho[\tau - t]} \left[\ln c(\tau) + \gamma \ln \frac{m(\tau)}{p(\tau)} \right] \mathrm{d}\tau.$$

$$\dot{a} = i[a - m] + w - T/L - pc.$$

6.4.2 Households: Optimality Conditions

- Money pays zero interest; capital earns the nominal return i.
- The nominal interest rate combines capital payments and asset-price changes.

$$i \equiv \frac{w^K + \dot{v}}{v}.$$

$$\frac{\dot{c}}{c} = i - \frac{\dot{p}}{p} - \rho.$$

$$m = \gamma \frac{pc}{i}.$$

6.4.3 Equilibrium: The Reduced Form

- Goods-market clearing requires total supply to equal $C + I$.
- With $\dot{K} = I - \delta K$, aggregate capital follows:

$$\dot{K} = Y(K, L) - C - \delta K. \tag{6.4.2}$$

- Since capital and consumption goods trade on the same market, $v = p$.

$$i = \frac{w^K}{p} + \frac{\dot{p}}{p} = \frac{\partial Y}{\partial K} + \frac{\dot{p}}{p}. \tag{6.4.3}$$

6.4.3 Equilibrium: Real and Nominal Rates

- The nominal interest rate equals the real interest rate plus inflation.
- Insert (6.4.3) into aggregate consumption growth.

$$\frac{\dot{C}}{C} = i - \frac{\dot{p}}{p} - \rho.$$

$$\frac{\dot{C}}{C} = \frac{\partial Y}{\partial K} - \rho. \tag{6.4.4}$$

real interest rate: $\dfrac{\partial Y}{\partial K}$, inflation: $\dfrac{\dot{p}}{p}$.

6.4.3 Equilibrium: Money Demand

- Aggregating households' money demand gives the money-market condition.
- Given an exogenous money supply rule, the reduced form determines K, C, i, p.

$$M = \gamma \frac{pC}{i}. \tag{6.4.5}$$

$$\begin{cases} \dot{K} = Y(K, L) - C - \delta K, \\ \dot{C}/C = \partial Y/\partial K - \rho, \\ i = \partial Y/\partial K + \dot{p}/p, \\ M = \gamma pC/i. \end{cases}$$

6.4.3 Equilibrium: Dichotomy

- With flexible prices, real variables are determined independently of monetary variables.
- Equations (6.4.2) and (6.4.4) determine K and C as in the optimal growth model without money.

$$\underbrace{(K, C)}_{\text{real side}} \quad \text{from} \quad \dot{K} = Y(K, L) - C - \delta K, \qquad \frac{\dot{C}}{C} = \frac{\partial Y}{\partial K} - \rho.$$

$$\underbrace{(p, i, M)}_{\text{nominal side}} \quad \text{from} \quad i = \frac{\partial Y}{\partial K} + \frac{\dot{p}}{p}, \qquad M = \gamma \frac{pC}{i}.$$

6.4.3 Equilibrium: A Steady State

- Suppose the technology implies constant long-run K, hence constant C.
- Substitute the nominal interest rate out of (6.4.3) and (6.4.5).

$$M = \gamma \frac{pC}{i} \quad \Rightarrow \quad i = \gamma \frac{pC}{M},$$
$$i = \frac{\partial Y}{\partial K} + \frac{\dot{p}}{p} \quad \Rightarrow \quad \frac{\dot{p}}{p} = \gamma \frac{pC}{M} - \frac{\partial Y}{\partial K}.$$

If M is constant, there exists p^* such that $\dot{p}/p = 0$.

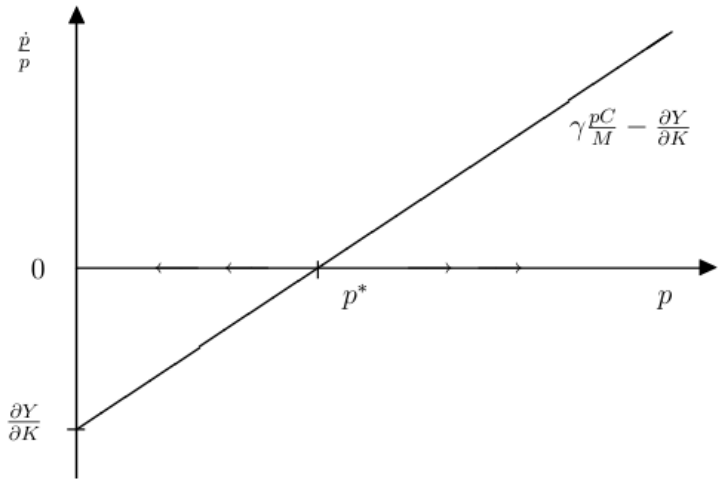


Figure 6.4.1 *The price level p^* in a monetary economy*

6.4.3 Equilibrium: Money Growth

- With constant GDP, money growth increases the long-run price level.
- For equilibria with constant inflation, (6.4.3) implies a constant nominal interest rate.

$$M = \gamma \frac{pC}{i}, \qquad C \text{ constant}, \qquad i \text{ constant}.$$

$$\frac{\dot{p}}{p} = \frac{\dot{M}}{M}. \tag{6.4.6}$$

Steady-state inflation

With constant GDP, inflation equals the growth rate of money supply.

6.4.3 Equilibrium: A Growth Equilibrium

- Suppose long-run real variables grow at exogenous rate g.
- Again assume a constant inflation rate, hence a constant nominal interest rate.

$$\frac{\dot{Y}}{Y} = \frac{\dot{C}}{C} = \frac{\dot{K}}{K} = g.$$

$$M = \gamma \frac{pC}{i}, \qquad i \text{ constant.}$$

$$\frac{\dot{p}}{p} = \frac{\dot{M}}{M} - \frac{\dot{C}}{C}.$$

Growth equilibrium inflation

Inflation equals money growth minus consumption growth.

6.4.3 Equilibrium: Exogenous Nominal Interest Rates

- The same reduced form can treat i as exogenous and M as endogenous.
- Flexible prices preserve the dichotomy: real quantities remain determined by (6.4.2) and (6.4.4).

Money-supply regime: M exogenous, i endogenous.

Interest-rate regime: i exogenous, M endogenous.

$$i = \frac{\partial Y}{\partial K} + \frac{\dot{p}}{p} \quad \Rightarrow \quad \frac{\dot{p}}{p} = i - \frac{\partial Y}{\partial K}.$$

6.4.3 Equilibrium: Interest-Rate Setting

- In this flexible-price setup, higher nominal interest rates imply higher inflation.
- The money-market equilibrium then determines the required money supply path.

$$\frac{\dot{p}}{p} = i - \frac{\partial Y}{\partial K}, \qquad M = \gamma \frac{pC}{i}.$$

Real side unchanged: $\dot{K} = Y(K, L) - C - \delta K, \qquad \frac{\dot{C}}{C} = \frac{\partial Y}{\partial K} - \rho.$